

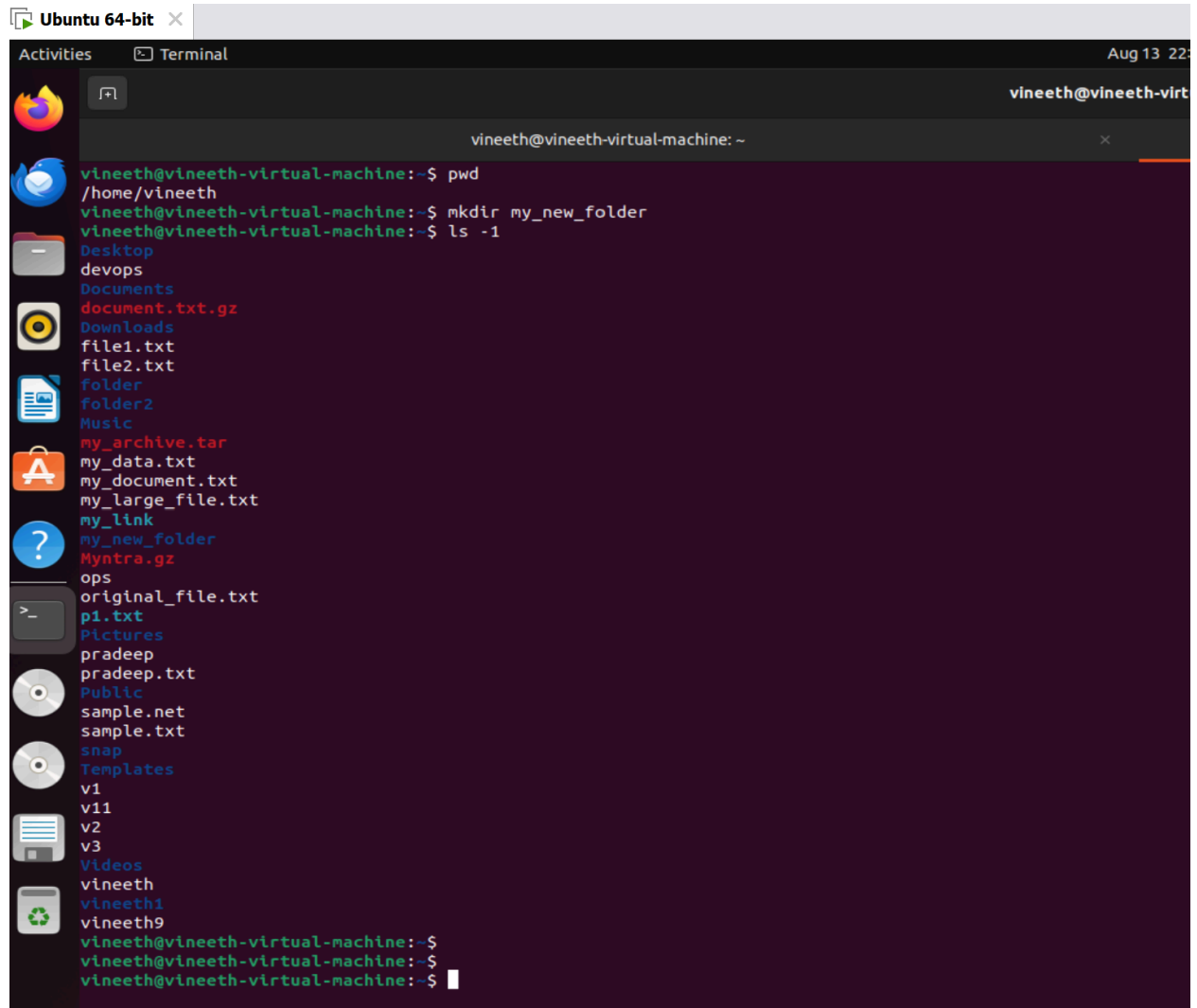
Create user with name Techie and provide sudo access to user.

```
vineeth@vineeth-virtual-machine: ~  
vineeth@vineeth-virtual-machine:~$ cd ~  
vineeth@vineeth-virtual-machine:~$ pwd  
/home/vineeth  
vineeth@vineeth-virtual-machine:~$ su root  
password:  
su: Authentication failure  
vineeth@vineeth-virtual-machine:~$ sudo su -  
sudo) password for vineeth:  
root@vineeth-virtual-machine:~# useradd techie  
root@vineeth-virtual-machine:~# passwd techie  
New password:  
BAD PASSWORD: The password is a palindrome  
retype new password:  
passwd: password updated successfully  
root@vineeth-virtual-machine:~# id techie  
uid=1003(techie) gid=1003(techie) groups=1003(techie)  
root@vineeth-virtual-machine:~# cd  
root@vineeth-virtual-machine:~#  
root@vineeth-virtual-machine:~# cd -  
root@vineeth-virtual-machine:~# su vineeth  
su: user vineeth does not exist or the user entry does not contain all the required fields  
root@vineeth-virtual-machine:~# sudo su vineeth  
vineeth@vineeth-virtual-machine:~$ cd  
vineeth@vineeth-virtual-machine:~$ touch pradeep.txt  
vineeth@vineeth-virtual-machine:~$ ls  
Desktop Documents Downloads file2.txt folder2 my_archive.tar my_document.txt my_link ops Pictures pradeep.txt sample.net snap v1 v2 vVideos vineeth1  
evops document.txt.gz file1.txt folder basic my_data.txt my_large_file.txt myntra.gz original_file.txt pradeep Public sample.txt Templates v11 v3 vineeth vineeth9  
vineeth@vineeth-virtual-machine:~$ vin pradeep.txt  
vineeth@vineeth-virtual-machine:~$ cat pi.txt  
hls tsssa  
vineeth@vineeth-virtual-machine:~$
```

Navigate to the home directory.

```
vineeth@vineeth-virtual-machine: ~  
vineeth@vineeth-virtual-machine:~$ pwd  
/home/vineeth  
vineeth@vineeth-virtual-machine:~$ cd ~  
vineeth@vineeth-virtual-machine:~$ cs ~  
Command 'cs' not found, but can be installed with:  
sudo apt install csound  
vineeth@vineeth-virtual-machine:~$ pwd  
/home/vineeth  
vineeth@vineeth-virtual-machine:~$
```

Create a new directory.

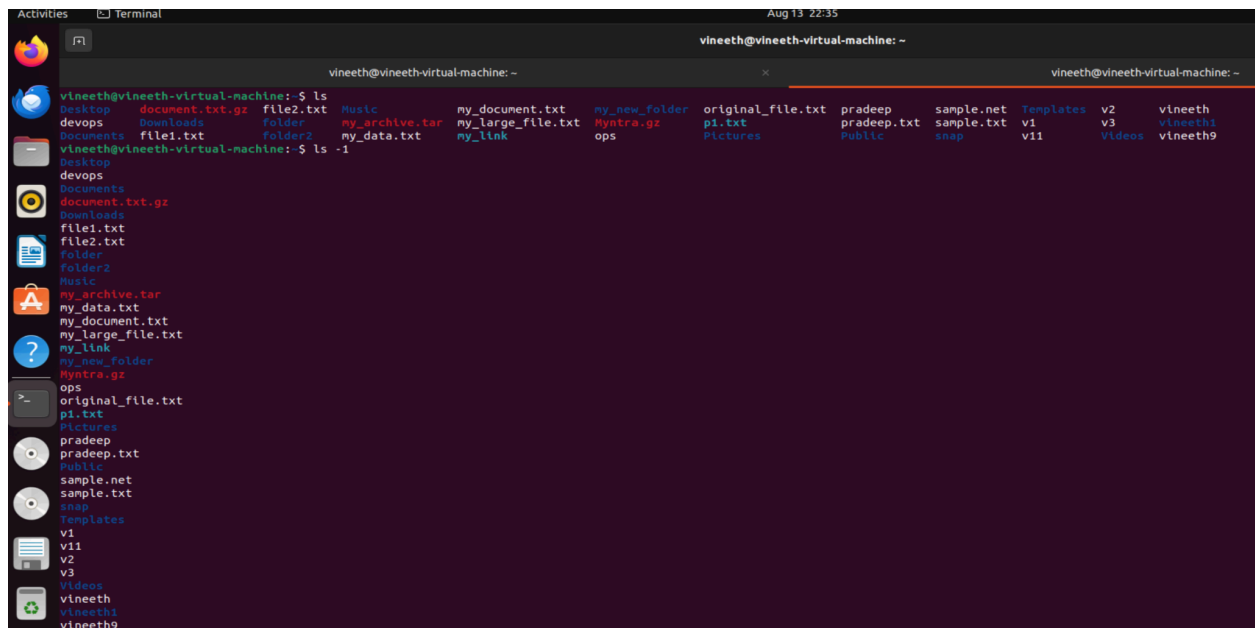


The screenshot shows a terminal window titled "Terminal" with the user "vineeth" on a "vineeth-virtual-machine". The terminal displays the following commands and output:

```
vineeth@vineeth-virtual-machine:~$ pwd
/home/vineeth
vineeth@vineeth-virtual-machine:~$ mkdir my_new_folder
vineeth@vineeth-virtual-machine:~$ ls -l
Desktop
devops
Documents
document.txt.gz
Downloads
file1.txt
file2.txt
folder
folder2
Music
my_archive.tar
my_data.txt
my_document.txt
my_large_file.txt
my_link
my_new_folder
Myntra.gz
ops
original_file.txt
p1.txt
Pictures
pradeep
pradeep.txt
Public
sample.net
sample.txt
snap
Templates
v1
v11
v2
v3
Videos
vineeth
vineeth1
vineeth9
vineeth@vineeth-virtual-machine:~$
vineeth@vineeth-virtual-machine:~$
vineeth@vineeth-virtual-machine:~$
```

The left sidebar of the terminal window shows a list of files and folders with their respective icons, including Desktop, devops, Documents, document.txt.gz, Downloads, file1.txt, file2.txt, folder, folder2, Music, my_archive.tar, my_data.txt, my_document.txt, my_large_file.txt, my_link, my_new_folder, Myntra.gz, ops, original_file.txt, p1.txt, Pictures, pradeep, pradeep.txt, Public, sample.net, sample.txt, snap, Templates, v1, v11, v2, v3, Videos, vineeth, vineeth1, and vineeth9.

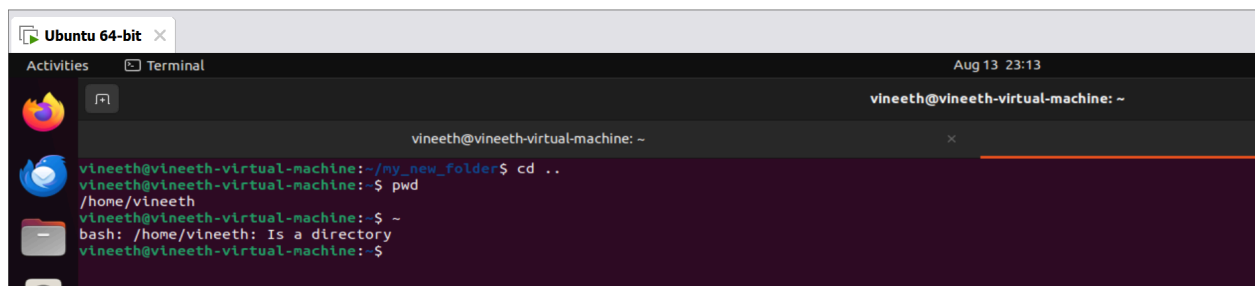
List the contents of a directory.



A terminal window titled 'vineeth@vineeth-virtual-machine: ~' showing the output of the 'ls' command. The output lists various files and directories in the current directory, including Desktop, devops, Downloads, Documents, file1.txt, file2.txt, folder, folder2, Music, my_archive.tar, my_data.txt, my_document.txt, my_large_file.txt, my_link, my_new_folder, myntra.gz, ops, original_file.txt, p1.txt, Pictures, pradeep, pradeep.txt, public, sample.net, sample.txt, snap, Templates, v1, v2, v3, Videos, vineeth, vineeth1, and vineeth9.

```
vineeth@vineeth-virtual-machine:~$ ls
Desktop  document.txt.gz  file2.txt  Music      my_document.txt  my_new_folder  original_file.txt  pradeep  sample.net  Templates  v2      vineeth
devops   Downloads        folder     my_archive.tar my_large_file.txt myntra.gz       p1.txt          pradeep.txt  sample.txt  v1        v3      vineeth1
Documents file1.txt        folder2    my_data.txt  my_link          ops            Pictures          public     snap       v11       v3      vineeth9
Desktop
devops
Documents
document.txt.gz
Downloads
file1.txt
file2.txt
folder
folder2
Music
my_archive.tar
my_data.txt
my_document.txt
my_large_file.txt
my_link
my_new_folder
myntra.gz
ops
original_file.txt
p1.txt
Pictures
pradeep
pradeep.txt
public
sample.net
sample.txt
snap
Templates
v1
v2
v3
Videos
vineeth
vineeth1
vineeth9
```

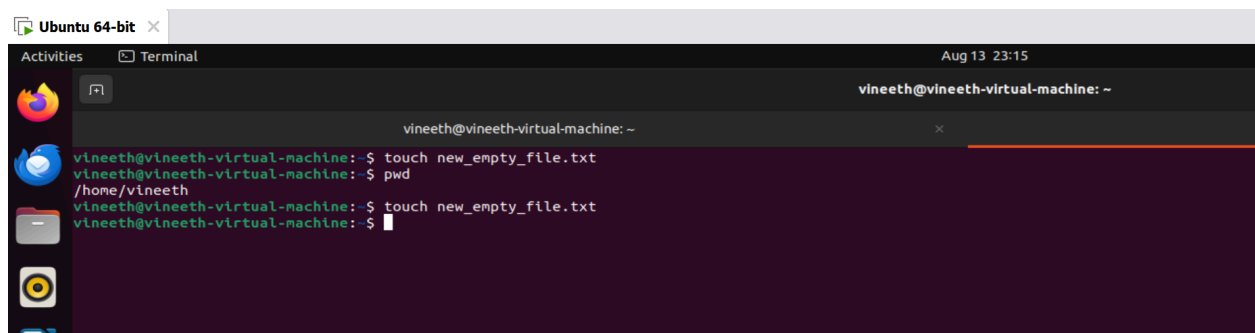
Change the current directory.



A terminal window titled 'vineeth@vineeth-virtual-machine: ~' showing the output of the 'cd ..' command. The prompt changes to '/home/vineeth'. The user then enters 'pwd', which outputs '/home/vineeth'. Finally, the user enters '~', which outputs 'bash: /home/vineeth: Is a directory' and then returns to the shell prompt.

```
vineeth@vineeth-virtual-machine:~$ cd ..
vineeth@vineeth-virtual-machine:~/home/vineeth$ pwd
/home/vineeth
vineeth@vineeth-virtual-machine:~/home/vineeth$ ~
bash: /home/vineeth: Is a directory
vineeth@vineeth-virtual-machine:~/home/vineeth$
```

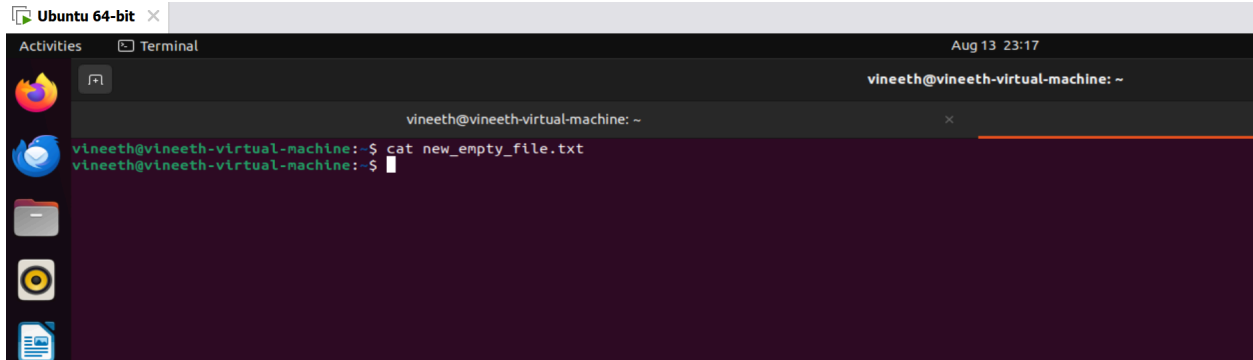
Create a new empty file.



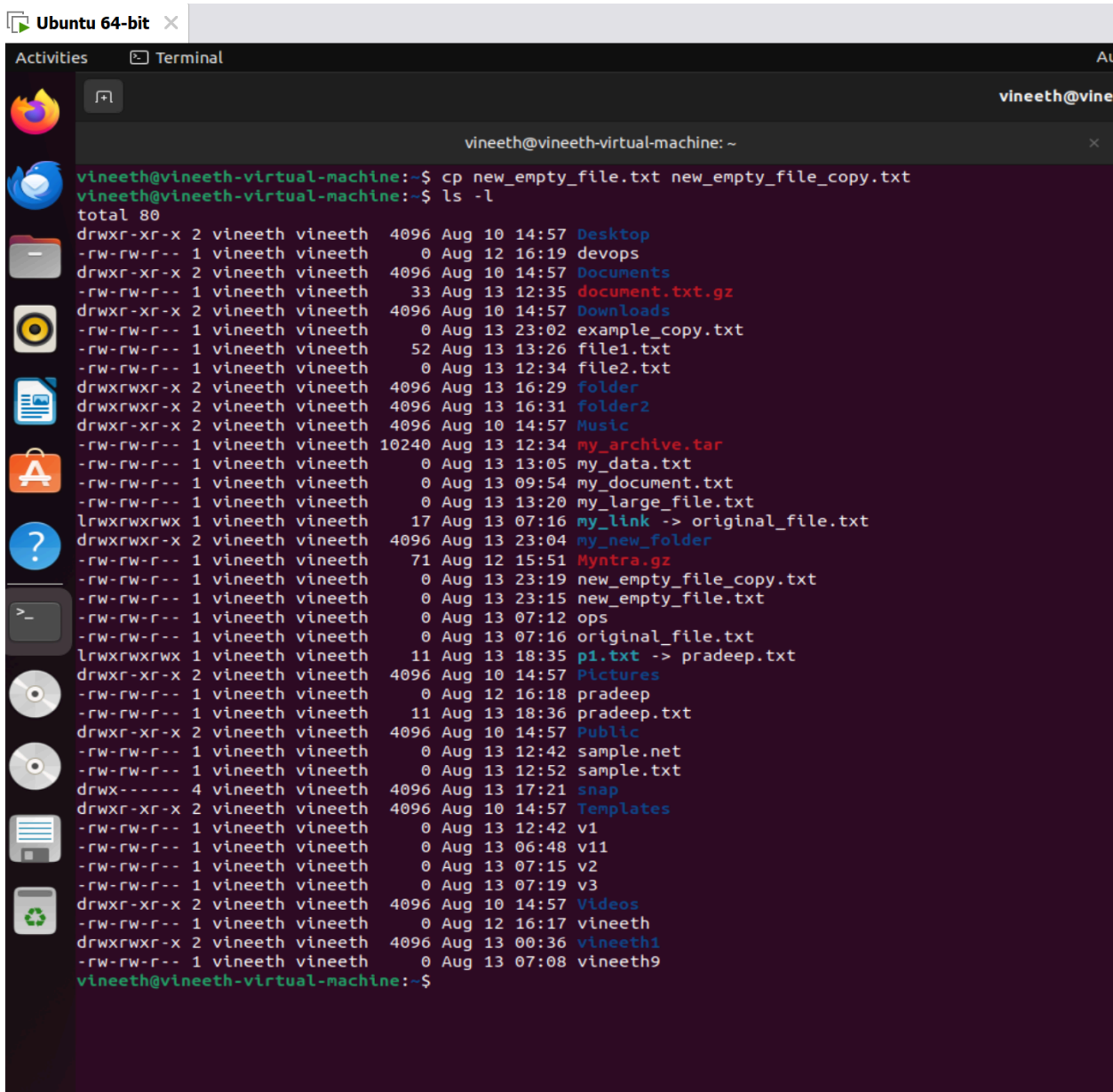
A terminal window titled 'vineeth@vineeth-virtual-machine: ~' showing the output of the 'touch' command. The user enters 'touch new_empty_file.txt', which creates a new empty file. The user then enters 'pwd', which outputs '/home/vineeth'. Finally, the user enters 'touch new_empty_file.txt' again, which does not change the output.

```
vineeth@vineeth-virtual-machine:~$ touch new_empty_file.txt
vineeth@vineeth-virtual-machine:~/home/vineeth$ pwd
/home/vineeth
vineeth@vineeth-virtual-machine:~/home/vineeth$ touch new_empty_file.txt
vineeth@vineeth-virtual-machine:~/home/vineeth$
```

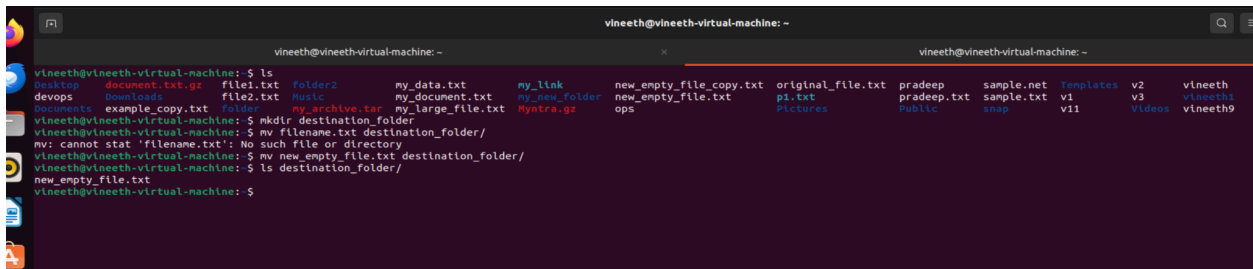
View the contents of a file.



Copy a file to another location.

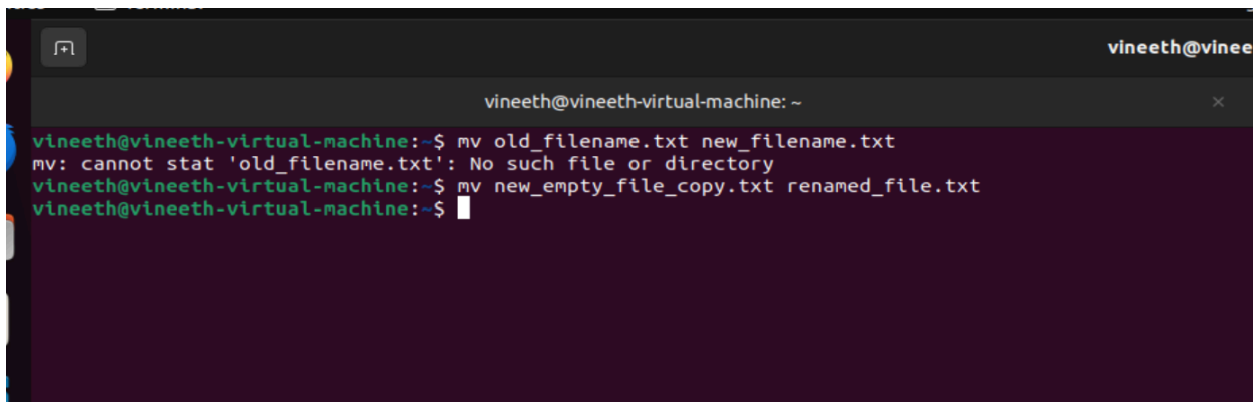


Move a file to another location.



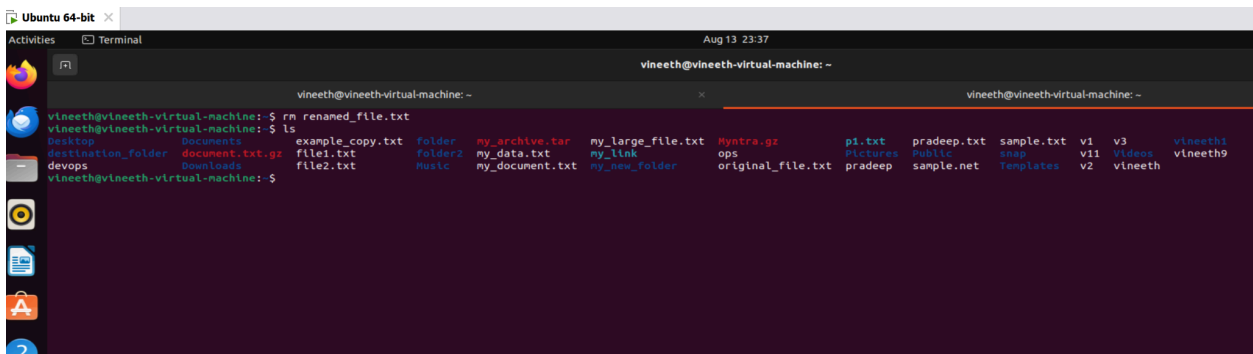
```
vineeth@vineeth-virtual-machine: ~  
vineeth@vineeth-virtual-machine:~$ ls  
Desktop  document.txt.gz  file1.txt  folder2  my_data.txt  my_link  new_empty_file_copy.txt  original_file.txt  pradeep  sample.net  Templates  v2  vineeth  
devops   Downloads  file2.txt  Music    my_document.txt  my_new_folder  new_empty_file.txt  pi.txt  pradeep.txt  sample.txt  v1  v3  vineeth9  
Documents  example_copy.txt  folder  my_archive.tar  my_large_file.txt  Nymtra.gz  ops  Pictures  Public  snap  v11  v2  vineeth9  
vineeth@vineeth-virtual-machine:~$ mkdir destination_folder  
vineeth@vineeth-virtual-machine:~$ mv filename.txt destination_folder/  
mv: cannot stat 'filename.txt': No such file or directory  
vineeth@vineeth-virtual-machine:~$ mv new_empty_file.txt destination_folder/  
vineeth@vineeth-virtual-machine:~$ ls destination_folder/  
new_empty_file.txt  
vineeth@vineeth-virtual-machine:~$
```

Rename a file.



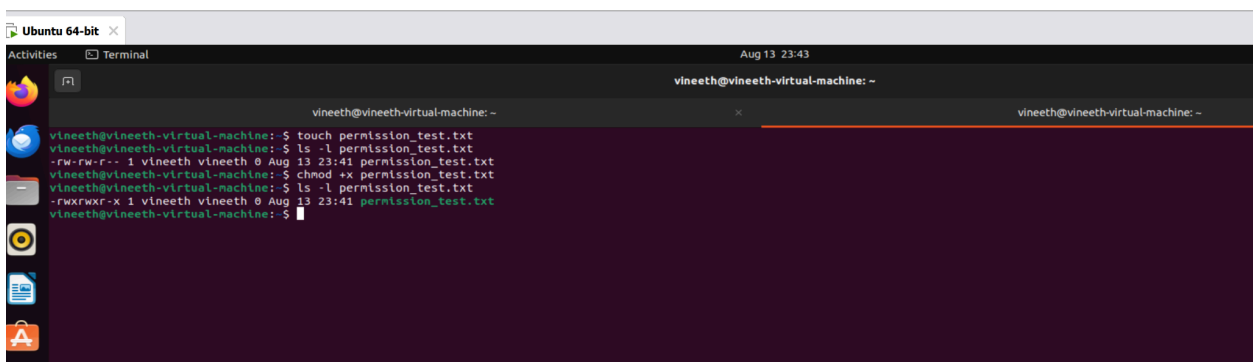
```
vineeth@vineeth-virtual-machine: ~  
vineeth@vineeth-virtual-machine:~$ mv old_filename.txt new_filename.txt  
mv: cannot stat 'old_filename.txt': No such file or directory  
vineeth@vineeth-virtual-machine:~$ mv new_empty_file_copy.txt renamed_file.txt  
vineeth@vineeth-virtual-machine:~$
```

Delete a file.



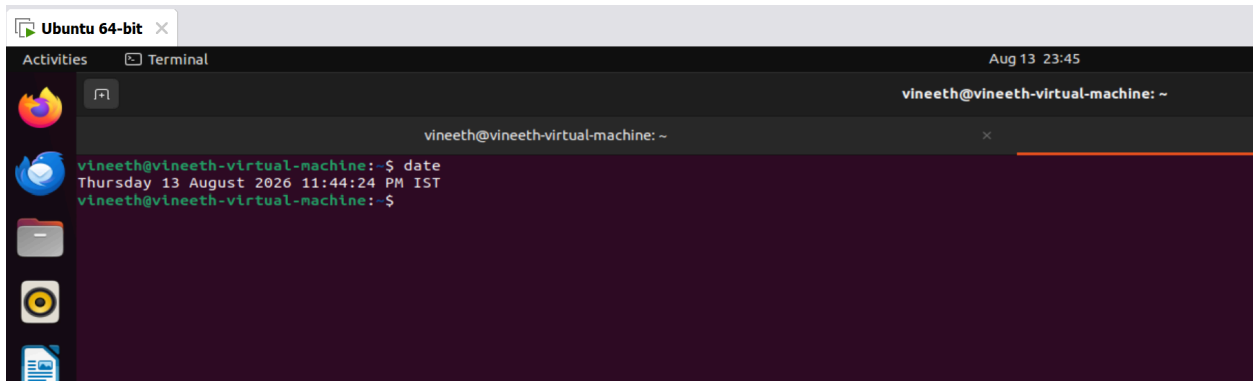
```
Ubuntu 64-bit x  
Activities Terminal Aug 13 23:37  
vineeth@vineeth-virtual-machine: ~  
vineeth@vineeth-virtual-machine:~$ rm renamed_file.txt  
vineeth@vineeth-virtual-machine:~$ ls  
Desktop  Documents  example_copy.txt  folder  my_archive.tar  my_large_file.txt  Nymtra.gz  pi.txt  pradeep.txt  sample.txt  v1  v3  vineeth1  
devops   Downloads  document.txt.gz  file1.txt  folder2  my_data.txt  my_link  ops  original_file.txt  pradeep  sample.net  snap  Templates  v11  v2  vineeth  
Documents  example_copy.txt  folder  my_archive.tar  my_large_file.txt  Nymtra.gz  ops  original_file.txt  pradeep  sample.net  snap  Templates  v11  v2  vineeth  
vineeth@vineeth-virtual-machine:~$
```

Grant or revoke permissions on a file or directory.

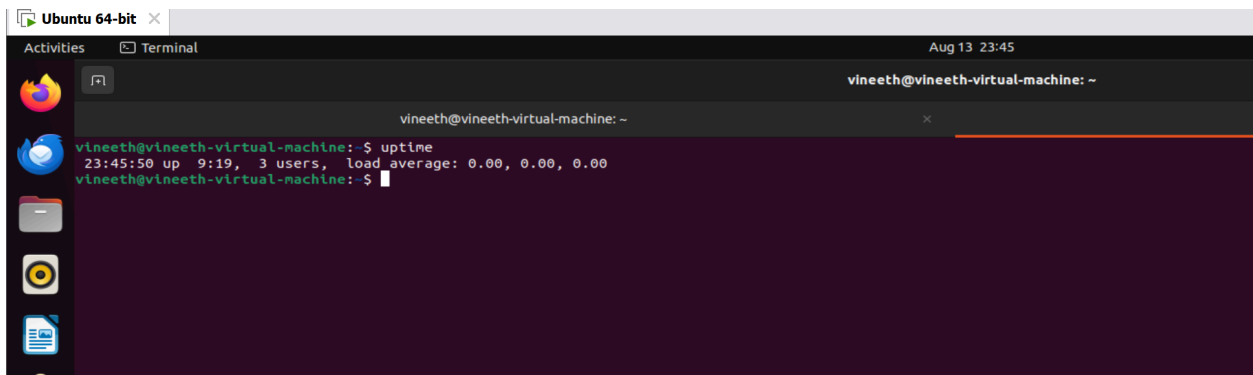


```
Ubuntu 64-bit x  
Activities Terminal Aug 13 23:43  
vineeth@vineeth-virtual-machine: ~  
vineeth@vineeth-virtual-machine:~$ touch permission_test.txt  
vineeth@vineeth-virtual-machine:~$ ls -l permission_test.txt  
-rw-rw-r-- 1 vineeth vineeth 0 Aug 13 23:41 permission_test.txt  
vineeth@vineeth-virtual-machine:~$ chmod +x permission_test.txt  
vineeth@vineeth-virtual-machine:~$ ls -l permission_test.txt  
-rwxrwxr-x 1 vineeth vineeth 0 Aug 13 23:41 permission_test.txt  
vineeth@vineeth-virtual-machine:~$
```

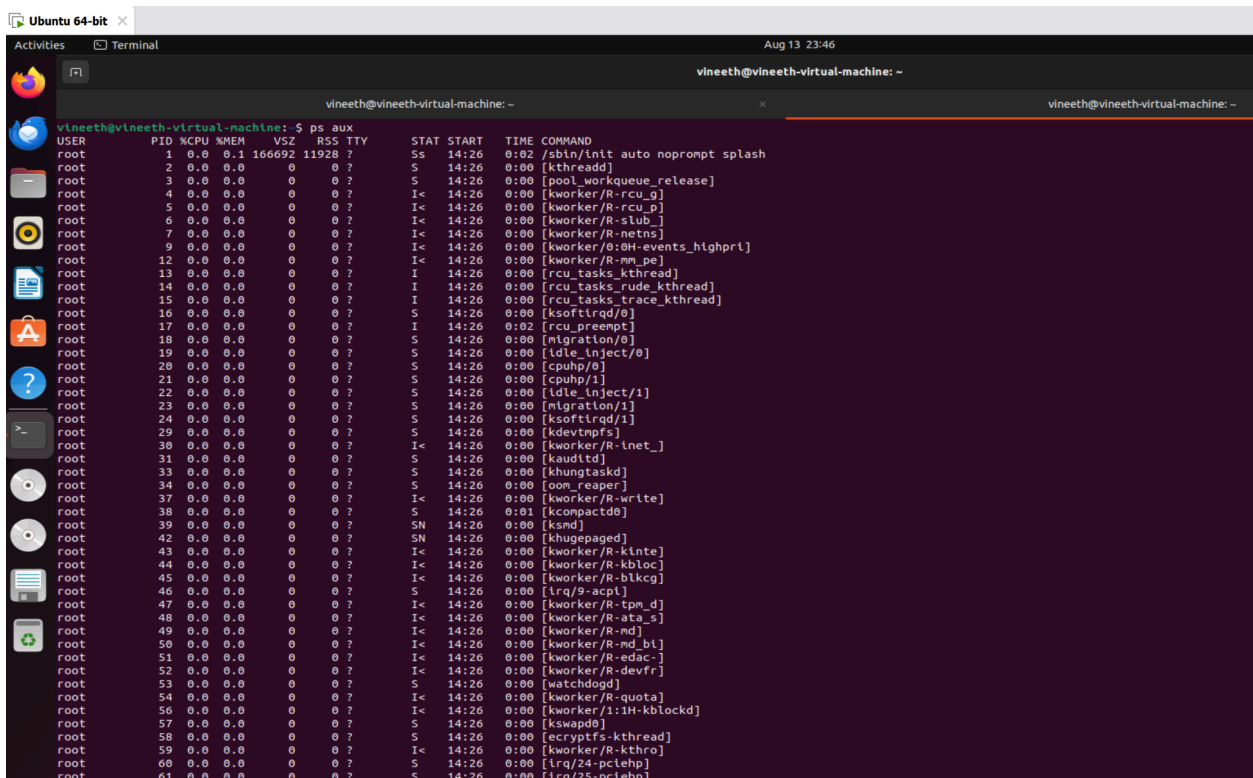
View the current date and time.



Check the system uptime.



View the running processes.



Kill a running process.

```
vineeth@vineeth-virtual-machine:~$ sleep 300 &
[1] 7481
vineeth@vineeth-virtual-machine:~$ kill 7481
vineeth@vineeth-virtual-machine:~$ ps aux | grep sleep
vineeth  7485  0.0  0.0  9080  2636 pts/3    R+   23:52   0:00 grep --color=auto sleep
[1]+  Terminated                  sleep 300
vineeth@vineeth-virtual-machine:~$
```

Install a package using the package manager (e.g., apt or yum).

```

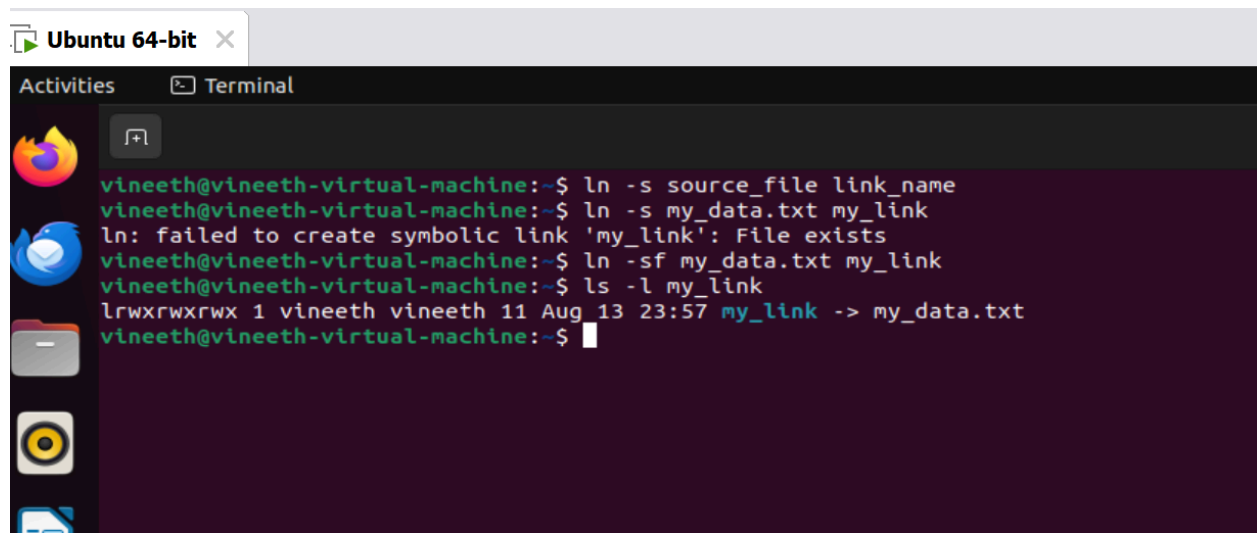
vineeth@vineeth-virtual-machine:~$ sudo apt update
[sudo] password for vineeth:
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadata [74.4 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [117 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 DEP-11 Metadata [360 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 DEP-11 Metadata [127 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-backports/main amd64 DEP-11 Metadata [5,788 B]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 DEP-11 Metadata [12.4 kB]
Fetched 1,082 kB in 3s (387 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
457 packages can be upgraded. Run 'apt list --upgradable' to see them.
vineeth@vineeth-virtual-machine:~$ htop
Command 'htop' not found, but can be installed with:
sudo apt install htop
vineeth@vineeth-virtual-machine:~$

```

Update the system packages.

[illegible]

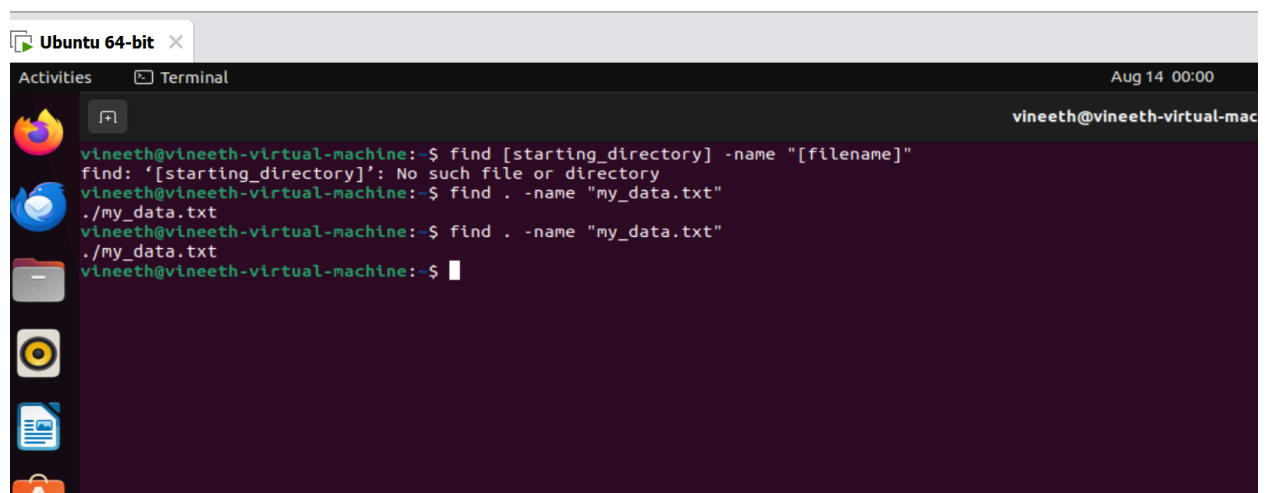
Create a symbolic link.



A terminal window titled 'Ubuntu 64-bit' with a dark background. The prompt is 'vineeth@vineeth-virtual-machine:~\$'. The user enters the command 'ln -s source_file link_name'. The prompt changes to 'vineeth@vineeth-virtual-machine:~\$' and the user enters 'ln -s my_data.txt my_link'. The terminal shows an error: 'ln: failed to create symbolic link 'my_link': File exists'. The user then enters 'ln -sf my_data.txt my_link'. The prompt changes to 'vineeth@vineeth-virtual-machine:~\$' and the user enters 'ls -l my_link'. The terminal shows the output: 'lrwxrwxrwx 1 vineeth vineeth 11 Aug 13 23:57 my_link -> my_data.txt'. The prompt returns to 'vineeth@vineeth-virtual-machine:~\$'.

```
vineeth@vineeth-virtual-machine:~$ ln -s source_file link_name
vineeth@vineeth-virtual-machine:~$ ln -s my_data.txt my_link
ln: failed to create symbolic link 'my_link': File exists
vineeth@vineeth-virtual-machine:~$ ln -sf my_data.txt my_link
vineeth@vineeth-virtual-machine:~$ ls -l my_link
lrwxrwxrwx 1 vineeth vineeth 11 Aug 13 23:57 my_link -> my_data.txt
vineeth@vineeth-virtual-machine:~$
```

Search for files using the find command.



A terminal window titled 'Ubuntu 64-bit' with a dark background. The prompt is 'vineeth@vineeth-virtual-machine:~\$'. The user enters the command 'find [starting_directory] -name "[filename]"'. The terminal shows the output: 'find: '[starting_directory]': No such file or directory'. The user then enters 'find . -name "my_data.txt"'. The terminal shows the output: './my_data.txt'. The user then enters 'find . -name "my_data.txt"'. The terminal shows the output: './my_data.txt'. The prompt returns to 'vineeth@vineeth-virtual-machine:~\$'.

```
vineeth@vineeth-virtual-machine:~$ find [starting_directory] -name "[filename]"
find: '[starting_directory]': No such file or directory
vineeth@vineeth-virtual-machine:~$ find . -name "my_data.txt"
./my_data.txt
vineeth@vineeth-virtual-machine:~$ find . -name "my_data.txt"
./my_data.txt
vineeth@vineeth-virtual-machine:~$
```

. Compress and decompress files using tar.

```

vineeth@vineeth-virtual-machine:~$ touch vineeth2txt
vineeth@vineeth-virtual-machine:~$ vi vineeth2txt
vineeth@vineeth-virtual-machine:~$ cat vineeth2txt
15
aug
vineeth@vineeth-virtual-machine:~$ tar -cvf vineeth3txt vineeth2txt
vineeth3txt
vineeth@vineeth-virtual-machine:~$ cat vineeth3txt
vineeth3txt000066400017500001750000000001015237670061013153 0ustar  vineethvineeth
15
aug
vineeth@vineeth-virtual-machine:~$ ls
desktop      document.txt.gz  file2.txt  Music      my_large_file.txt  ops      Pictures      sample.net  v1  Videos      vineeth2txt
destination_folder  Downloads      folder      my_archive.tar  my_link      original_file.txt  pradeep    sample.txt  v11 vineeth      vineeth3txt
devops       example_copy.txt  folder2     my_data.txt    my_new_folder  pi.txt      pradeep.txt  snap       v2  vineeth1     vineeth9
documents    file1.txt         link_name   my_document.txt  Myntra.gz     permission_test.txt  Public     Templates  v3  vineeth1txt  vineethtxt
vineeth@vineeth-virtual-machine:~$ touch pradeep.txt
vineeth@vineeth-virtual-machine:~$ vi pradeep.txt
vineeth@vineeth-virtual-machine:~$ cat pradeep.txt
hl
:Venuria
vineeth@vineeth-virtual-machine:~$ tar -czvf vineethnew.gz pradeep.txt
pradeep.txt
vineeth@vineeth-virtual-machine:~$ cat vineethnew
cat: vineethnew: No such file or directory
vineeth@vineeth-virtual-machine:~$ ls
desktop      document.txt.gz  file2.txt  Music      my_large_file.txt  ops      Pictures      sample.net  v1  Videos      vineeth2txt  vineethtxt
destination_folder  Downloads      folder      my_archive.tar  my_link      original_file.txt  pradeep    sample.txt  v11 vineeth      vineeth3txt
devops       example_copy.txt  folder2     my_data.txt    my_new_folder  pi.txt      pradeep.txt  snap       v2  vineeth1     vineeth9
documents    file1.txt         link_name   my_document.txt  Myntra.gz     permission_test.txt  Public     Templates  v3  vineeth1txt  vineethnew.gz
vineeth@vineeth-virtual-machine:~$ touch jashtxt
vineeth@vineeth-virtual-machine:~$ vi jashtxt
vineeth@vineeth-virtual-machine:~$ cat jashtxt
duke
vineeth@vineeth-virtual-machine:~$ tar -czvf jashnew.gz jashtxt
jashtxt
vineeth@vineeth-virtual-machine:~$ cat jashtxt
duke
vineeth@vineeth-virtual-machine:~$ ls
desktop      document.txt.gz  file2.txt  jashtxt      my_data.txt    my_new_folder  pi.txt      pradeep.txt  snap       v2  vineeth1     vineeth9
destination_folder  Downloads      folder      link_name     my_document.txt  Myntra.gz     permission_test.txt  Public     Templates  v3  vineeth1txt  vineethnew.gz
devops       example_copy.txt  folder2     Music      my_large_file.txt  ops      Pictures      sample.net  v1  Videos      vineeth2txt  vineethtxt
documents    file1.txt         jashnew.gz  my_archive.tar  my_link      original_file.txt  pradeep    sample.txt  v11 vineeth      vineeth3txt
vineeth@vineeth-virtual-machine:~$

```

22 Monitor system resources with top or htop.

top

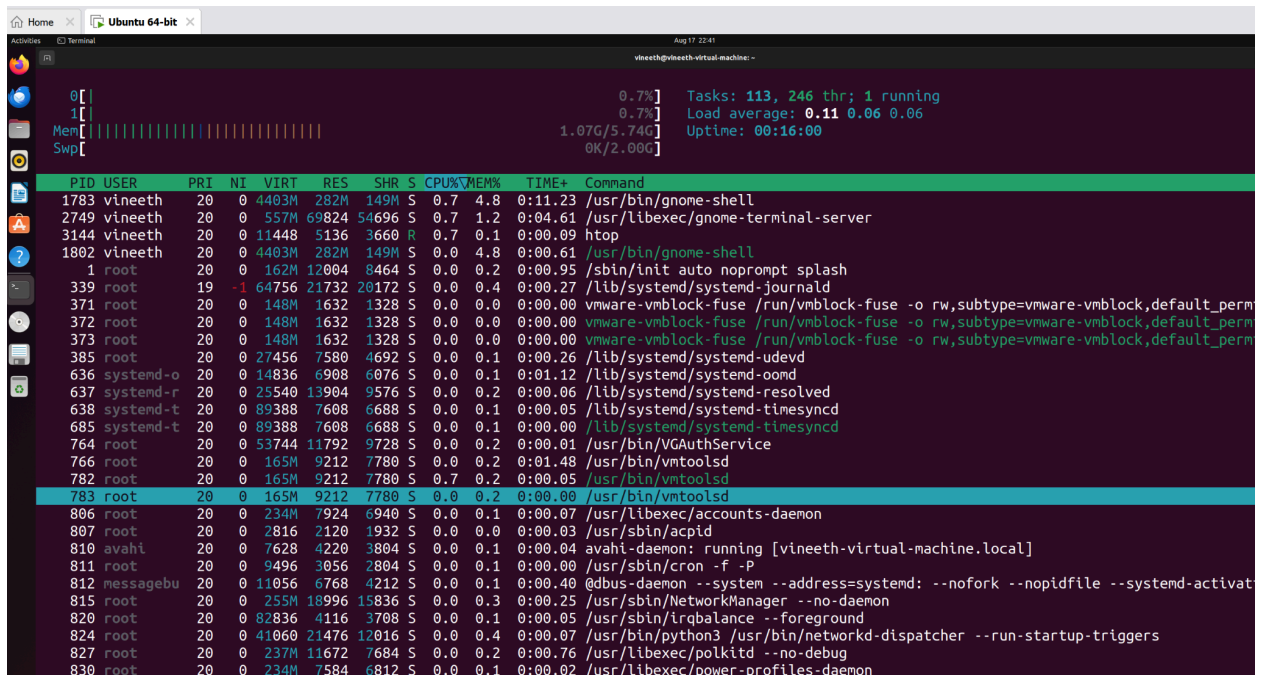
```

top - 22:35:54 up 10 min, 1 user, load average: 0.04, 0.08, 0.07
Tasks: 292 total, 1 running, 291 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.0 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 5878.4 total, 3689.7 free, 1007.1 used, 1181.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 4573.4 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
 269 root        20   0      0      0      0   I   0.3   0.0   0:00.12 kworker/u257:28-events_freezable_power_
1783 vineeth    20   0 4230148 288208 153196 S   0.3   4.8   0:08.79 gnome-shell
2749 vineeth    20   0 569872 69228 54116 S   0.3   1.2   0:03.15 gnome-terminal-
2890 vineeth    20   0 13228 4336 3400 R   0.3   0.1   0:00.03 top
    1 root        20   0 166676 12004 8464 S   0.0   0.2   0:23.02 systemd
    2 root        20   0      0      0      0   S   0.0   0.0   0:00.01 kthreadd
    3 root        20   0      0      0      0   S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/R-rcu_g
    5 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/R-rcu_p
    6 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/R-slub_
    7 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/R-netns
    9 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
   11 root        20   0      0      0      0   I   0.0   0.0   0:00.00 kworker/u256:0-ext4-rsv-conversion
   12 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/R-mm_pe
   13 root        20   0      0      0      0   I   0.0   0.0   0:00.00 rcu_tasks_kthread
   14 root        20   0      0      0      0   I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   15 root        20   0      0      0      0   I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   16 root        20   0      0      0      0   S   0.0   0.0   0:00.04 ksoftirqd/0
   17 root        20   0      0      0      0   I   0.0   0.0   0:00.14 rcu_preempt
   18 root        rt    0      0      0      0   S   0.0   0.0   0:00.00 migration/0
   19 root       -51   0      0      0      0   S   0.0   0.0   0:00.00 idle_inject/0
   20 root        20   0      0      0      0   S   0.0   0.0   0:00.00 cpuhp/0
   21 root        20   0      0      0      0   S   0.0   0.0   0:00.00 cpuhp/1
   22 root       -51   0      0      0      0   S   0.0   0.0   0:00.00 idle_inject/1
   23 root        rt    0      0      0      0   S   0.0   0.0   0:00.20 migration/1
   24 root        20   0      0      0      0   S   0.0   0.0   0:00.04 ksoftirqd/1
   25 root        20   0      0      0      0   I   0.0   0.0   0:00.27 kworker/1:0-events
   26 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
   29 root        20   0      0      0      0   S   0.0   0.0   0:00.00 kdevtmpfs
   30 root         0 -20      0      0      0   I   0.0   0.0   0:00.00 kworker/R-inet_

```

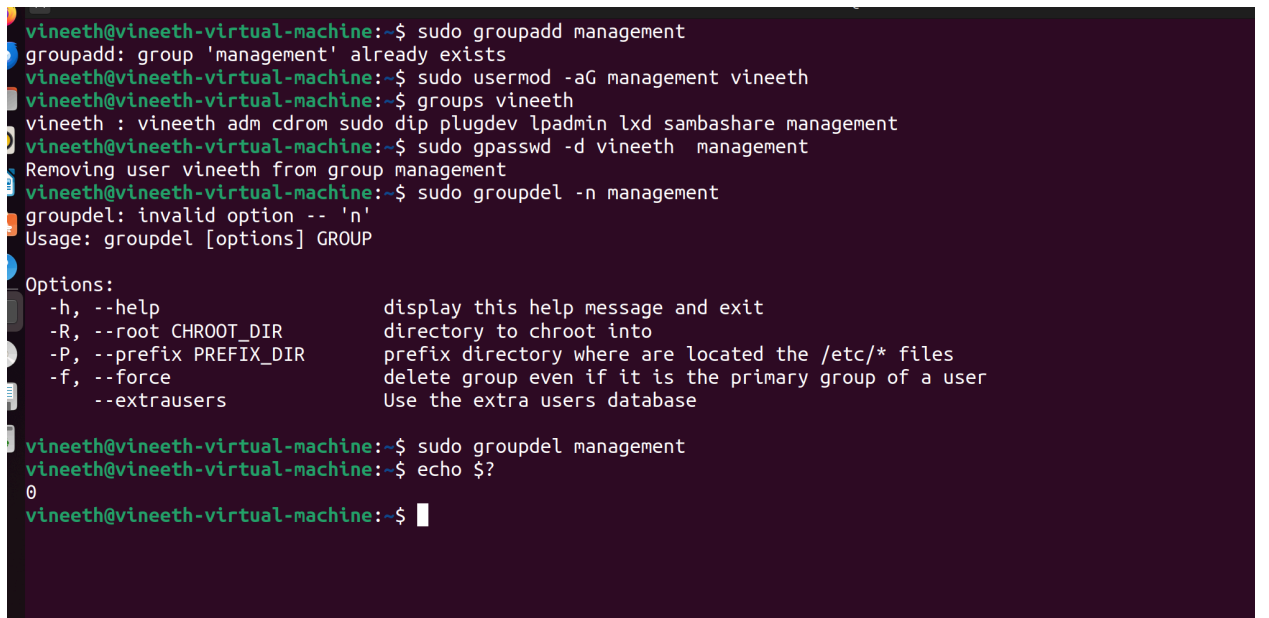
h top



```
0[ 0.7%] Tasks: 113, 246 thr; 1 running
1[ 0.7%] Load average: 0.11 0.06 0.06
Mem[ 1.07G/5.74G] Uptime: 00:16:00
Swp[ 0K/2.00G]

  PID USER   PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command
 1783 vineeth 20    0 4403M 282M 149M  S   0.7  4.8  0:11.23 /usr/bin/gnome-shell
 2749 vineeth 20    0 557M 69824 54696  S   0.7  1.2  0:04.61 /usr/libexec/gnome-terminal-server
 3144 vineeth 20    0 11448 5136 3660  R   0.7  0.1  0:00.09 htop
 1802 vineeth 20    0 4403M 282M 149M  S   0.0  4.8  0:00.61 /usr/bin/gnome-shell
    1 root    20    0 162M 12004 8464  S   0.0  0.2  0:00.95 /sbin/init auto noprompt splash
 339 root    19   -1 64756 21732 20172  S   0.0  0.4  0:00.27 /lib/systemd/systemd-journald
 371 root    20    0 148M 1632 1328  S   0.0  0.0  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_perm
 372 root    20    0 148M 1632 1328  S   0.0  0.0  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_perm
 373 root    20    0 148M 1632 1328  S   0.0  0.0  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_perm
 385 root    20    0 27456 7580 4692  S   0.0  0.1  0:00.26 /lib/systemd/systemd-udevd
 636 systemd-o 20    0 14836 6908 6076  S   0.0  0.1  0:01.12 /lib/systemd/systemd-oomd
 637 systemd-r 20    0 25540 13904 9576  S   0.0  0.2  0:00.06 /lib/systemd/systemd-resolved
 638 systemd-t 20    0 89388 7608 6688  S   0.0  0.1  0:00.05 /lib/systemd/systemd-timesyncd
 685 systemd-t 20    0 89388 7608 6688  S   0.0  0.1  0:00.00 /lib/systemd/systemd-timesyncd
 764 root    20    0 53744 11792 9728  S   0.0  0.2  0:00.01 /usr/bin/VGAuthService
 766 root    20    0 165M 9212 7780  S   0.0  0.2  0:01.48 /usr/bin/vmtoolsd
 782 root    20    0 165M 9212 7780  S   0.7  0.2  0:00.05 /usr/bin/vmtoolsd
 783 root    20    0 165M 9212 7780  S   0.0  0.2  0:00.00 /usr/bin/vmtoolsd
 806 root    20    0 234M 7924 6940  S   0.0  0.1  0:00.07 /usr/libexec/accounts-daemon
 807 root    20    0 2816 2120 1932  S   0.0  0.0  0:00.03 /usr/sbin/acpid
 810 avahi    20    0 7628 4220 3804  S   0.0  0.1  0:00.04 avahi-daemon: running [vineeth-virtual-machine.local]
 811 root    20    0 9496 3056 2804  S   0.0  0.1  0:00.00 /usr/sbin/cron -f -P
 812 messagebu 20    0 11056 6768 4212  S   0.0  0.1  0:00.40 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activat
 815 root    20    0 255M 18996 15836  S   0.0  0.3  0:00.25 /usr/sbin/NetworkManager --no-daemon
 820 root    20    0 82836 4116 3708  S   0.0  0.1  0:00.05 /usr/sbin/iqbalance --foreground
 824 root    20    0 41060 21476 12016  S   0.0  0.4  0:00.07 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
 827 root    20    0 237M 11672 7684  S   0.0  0.2  0:00.76 /usr/libexec/polkitd --no-debug
 830 root    20    0 234M 7584 6812  S   0.0  0.1  0:00.02 /usr/libexec/power-profiles-daemon
```

Create and manage user groups.



```
vineeth@vineeth-virtual-machine:~$ sudo groupadd management
groupadd: group 'management' already exists
vineeth@vineeth-virtual-machine:~$ sudo usermod -aG management vineeth
vineeth@vineeth-virtual-machine:~$ groups vineeth
vineeth : vineeth adm cdrom sudo dip plugdev lpadmin lxd sambashare management
vineeth@vineeth-virtual-machine:~$ sudo gpasswd -d vineeth management
Removing user vineeth from group management
vineeth@vineeth-virtual-machine:~$ sudo groupdel -n management
groupdel: invalid option -- 'n'
Usage: groupdel [options] GROUP

Options:
  -h, --help                display this help message and exit
  -R, --root CHROOT_DIR     directory to chroot into
  -P, --prefix PREFIX_DIR   prefix directory where are located the /etc/* files
  -f, --force                delete group even if it is the primary group of a user
      --extrausers            Use the extra users database

vineeth@vineeth-virtual-machine:~$ sudo groupdel management
vineeth@vineeth-virtual-machine:~$ echo $?
0
vineeth@vineeth-virtual-machine:~$
```

Monitor log files using tail or grep.

```

vineeth@vineeth-virtual-machine:~$ tail /var/log/syslog
Aug 17 23:34:38 vineeth-virtual-machine NetworkManager[815]: <info> [1786989878.9448] manager: NetworkManager state is now CONNECTED_SITE
Aug 17 23:34:38 vineeth-virtual-machine NetworkManager[815]: <info> [1786989878.9448] policy: set 'Wired connection 1' (ens33) as default for IPv4 routing
and DNS
Aug 17 23:34:38 vineeth-virtual-machine NetworkManager[815]: <info> [1786989878.9457] device (ens33): Activation: successful, device activated.
Aug 17 23:34:38 vineeth-virtual-machine systemd-resolved[637]: ens33: Bus client set search domain list to: localdomain
Aug 17 23:34:38 vineeth-virtual-machine systemd-resolved[637]: ens33: Bus client set default route setting: yes
Aug 17 23:34:38 vineeth-virtual-machine systemd-resolved[637]: ens33: Bus client set DNS server list to: 192.168.255.2
Aug 17 23:34:39 vineeth-virtual-machine NetworkManager[815]: <info> [1786989879.5853] manager: NetworkManager state is now CONNECTED_GLOBAL
Aug 17 23:34:45 vineeth-virtual-machine systemd-resolved[637]: Using degraded feature set UDP instead of UDP+EDNS0 for DNS server 192.168.255.2.
Aug 17 23:34:49 vineeth-virtual-machine systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
Aug 17 23:49:38 vineeth-virtual-machine NetworkManager[815]: <info> [1786990778.9419] dhcp4 (ens33): state changed new lease, address=192.168.255.128
vineeth@vineeth-virtual-machine:~$

```

Set up a web server (e.g., Apache or Nginx).

```

vineeth@vineeth-virtual-machine:~$ sudo apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
450 packages can be upgraded. Run 'apt list --upgradable' to see them.
vineeth@vineeth-virtual-machine:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.18.0-6ubuntu14.18).
0 upgraded, 0 newly installed, 0 to remove and 450 not upgraded.
vineeth@vineeth-virtual-machine:~$ sudo ufw allow 'Nginx HTTP'
Skipping adding existing rule
Skipping adding existing rule (v6)
vineeth@vineeth-virtual-machine:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-17 23:59:57 IST; 5min ago
     Docs: man:nginx(8)
   Process: 4528 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 4529 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Main PID: 4620 (nginx)
    Tasks: 3 (limit: 6960)
   Memory: 3.9M
      CPU: 46ms
   CGroup: /system.slice/nginx.service
           └─4620 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─4622 "nginx: worker process"
               └─4623 "nginx: worker process"

Aug 17 23:59:57 vineeth-virtual-machine systemd[1]: Starting A high performance web server and a reverse proxy server...
Aug 17 23:59:57 vineeth-virtual-machine systemd[1]: Started A high performance web server and a reverse proxy server.
vineeth@vineeth-virtual-machine:~$ curl -I http://localhost
Command 'curl' not found, but can be installed with:
sudo apt install curl
vineeth@vineeth-virtual-machine:~$

```

Set up a Application Server (e.g.,Apache Tomcat)

```

update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/rmic to provide /usr/bin/rmic (rmic) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jaotc to provide /usr/bin/jaotc (jaotc) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up default-jre (2:1.11-72build2) ...
Setting up default-jdk-headless (2:1.11-72build2) ...
Setting up openjdk-11-jdk:amd64 (11.0.31+11-1ubuntu1-22.04.2) ...
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Setting up default-jdk (2:1.11-72build2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ca-certificates (20230311ubuntu0.22.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
done.
Processing triggers for sgml-base (1.30) ...
Processing triggers for mailcap (3.70+nm1ubuntu1) ...
Setting up x11proto-dev (2021.5-1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for desktop-file-utils (0.26-1ubuntu3) ...
Setting up libxau-dev:amd64 (1:1.0.9-1build5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Setting up libice-dev:amd64 (2:1.0.10-1build2) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Setting up libsm-dev:amd64 (2:1.2.3-1build2) ...
Setting up libxdmcp-dev:amd64 (1:1.1.3-0ubuntu5) ...
Setting up libxcb1-dev:amd64 (1.14-3ubuntu3) ...
Setting up libx11-dev:amd64 (2:1.7.5-1ubuntu0.3) ...
Setting up libxt-dev:amd64 (1:1.2.1-1) ...
vineeth@vineeth-virtual-machine:~$ sudo groupadd tomcat
sudo useradd -s /bin/false -g tomcat -d /opt/tomcat tomcat
vineeth@vineeth-virtual-machine:~$ cd /tmp
curl -O https://downloads.apache.org/tomcat/tomcat-10/v10.1.24/bin/apache-tomcat-10.1.24.tar.gz
sudo mkdir /opt/tomcat
sudo tar xzvf apache-tomcat-*.tar.gz -C /opt/tomcat --strip-components=1
Command 'curl' not found, but can be installed with:
sudo apt install curl
tar (child): apache-tomcat-*.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
vineeth@vineeth-virtual-machine:~$ sudo chown -R tomcat:tomcat /opt/tomcat
sudo chmod +x /opt/tomcat/bin/*.sh
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
vineeth@vineeth-virtual-machine:~$ sudo systemctl daemon-reload
sudo systemctl start tomcat
sudo systemctl enable tomcat
Failed to start tomcat.service: Unit tomcat.service not found.
Failed to enable unit: Unit file tomcat.service does not exist.
vineeth@vineeth-virtual-machine:~$

```